



REDD PANTANAL



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CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Audit History.....	5
1.3	Sectoral Scope and Project Type	5
1.4	Project Eligibility	5
1.5	Project Design	6
1.6	Project Proponent	8
1.7	Other Entities Involved in the Project	8
1.8	Ownership.....	9
1.9	Project Start Date	9
1.10	Project Crediting Period	9
1.11	Project Scale and Estimated GHG Emission Reductions or Removals	10
1.12	Description of the Project Activity	12
1.13	Project Location	15
1.14	Conditions Prior to Project Initiation	15
1.15	Compliance with Laws, Statutes and Other Regulatory Frameworks	16
1.16	Double Counting and Participation under Other GHG Programs	21
1.17	Double Claiming, Other Forms of Credit, and Scope 3 Emissions.....	22
1.18	Sustainable Development Contributions	23
1.19	Additional Information Relevant to the Project	24
2	SAFEGUARDS AND STAKEHOLDER ENGAGEMENT	24
2.1	Stakeholder Engagement and Consultation.....	24
2.2	Risks to Stakeholders and the Environment.....	25
2.3	Respect for Human Rights and Equity	25
2.4	Ecosystem Health	25
3	APPLICATION OF METHODOLOGY.....	25
3.1	Title and Reference of Methodology	25
3.2	Applicability of Methodology	27
3.3	Project Boundary	34
3.4	Baseline Scenario	34

3.5	Additionality	34
3.6	Methodology Deviations	34
4	QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS.....	34
4.1	Baseline Emissions	34
4.2	Project Emissions	34
4.3	Leakage Emissions	34
4.4	Estimated GHG Emission Reductions and Carbon Dioxide Removals	34
5	MONITORING	35
5.1	Data and Parameters Available at Validation	35
5.2	Data and Parameters Monitored.....	35
5.3	Monitoring Plan.....	35
	APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION.....	36
	APPENDIX X: <TITLE OF APPENDIX>	36

1 PROJECT DETAILS

1.1 Summary Description of the Project

REDD Pantanal is a multi-activity grouped project located in Brazil, in the Pantanal biome. The project is classified under scope 14 - AFOLU (Agriculture, Forestry, and Other Land Uses) and falls into the category of Reducing Emissions from Deforestation and Degradation (REDD) in the scopes of planned (APDef) and unplanned (AUDef) deforestation.

The main goal of REDD Pantanal is to reduce global greenhouse gas (GHG) emissions by conserving native vegetation in the Pantanal biome. The project is based on the conservation of the remaining forest cover of properties under pressure from deforestation and forest degradation.

Information reported in the Annual Deforestation Report of Brazil¹, conducted by MapBiomas, on the monitoring of forest cover in the Pantanal biome, indicates that the suppression of native vegetation in the Pantanal has reached elevated levels in recent years. The annual deforestation rate in the biome was 31,211.7 hectares in 2022 (January 1, 2022, to December 31, 2022), with an average deforestation speed of 85.5 hectares per day, resulting in an 8.2% increase compared to 2021.

The increases in deforestation observed in 2022 in the Pantanal corroborate the advancement of deforestation activity in natural forests in Brazil, placing us at a critical juncture of imbalance in natural systems, affecting provisions of energy, water, and food security in Brazil and worldwide.

Given the high rates of deforestation in this biome, the project seeks landowners who have properties in the Pantanal and choose not to deforest their properties within legally permitted portions (APDef). Additionally, the project aims to ensure the protection of Permanent Preservation Areas (APP) and Legal Reserves (RL) on their properties (AUDef), which have also been targeted by illegal deforestation and degradation in recent decades.

In this context, ECCON, as the project proponent, will be responsible for the development of all stages, implementation across different instances, and monitoring of the climatic parameters of REDD Pantanal. The other entities involved in the project are the landowners, who, through the signing of a contract and letter of intent, attest to their commitment to maintaining the areas of their properties in a conserved state in the long term. This commitment allows them to support the activities to be developed by the project.

It is important to highlight that the revenue from carbon credits generated by REDD Pantanal will contribute to conservation, making the activity more appealing for the rural producers supported by the project. Therefore, maintaining the "standing forest" will be a competitive alternative to other land uses that pose a threat to the conversion of native vegetation in the current scenario (i.e., agricultural or livestock production).

¹ Mapbiomas Summary Deforestation report in [English](#) and Full report in [Portuguese](#).

REDD Pantanal activities commenced in 2021, with a 45-year credit period, from November 18th, 2021, to November 17th, 2066. PAI 01 is the first instance of the project. It will prevent the emission of 1,837,304 tCO₂e over 30 years, with an average annual reduction of 61,243 tCO₂e).

1.2 Audit History

This section does not apply to this project at its current stage of development.

1.3 Sectoral Scope and Project Type

Sectoral scope	14 – Agriculture, Forestry, and Other Land Use (AFOLU)
AFOLU project category²	Reducing Emissions from Deforestation and Degradation (REDD)
Project activity type	Avoiding Unplanned Deforestation (AUDef) Avoiding Planned Deforestation (APDef)

1.4 Project Eligibility

1.4.1 General eligibility

The REDD Pantanal is a grouped project, designed to include more than one Project Activity Instance ("PAI") through the inclusion of new areas that meet all the proposed eligibility criteria, which allows the expansion of project activities throughout its development.

As detailed in section 3.2, the project meets the eligibility criteria outlined in VM0007, which is an approved methodology under the VCS Program. Additionally, none of the project activities are excluded according to VCS standard (v4.5, Table 1, pg. 2). Consequently, this project is fully included within the scope of the VCS Program.

1.4.2 AFOLU project eligibility

According to VCS v4.5, for AFOLU projects, eligible REDD activities are those that reduce net GHG emissions by reducing deforestation and/or degradation of forests. The project area shall meet an internationally accepted definition of forest, such as those based on UNFCCC host country thresholds or FAO definitions and shall qualify as forest for a minimum of 10 years before the project start date. The definition of forest may include mature forests, secondary forests, and degraded forests. Under the VCS Program, secondary forests are forests that (i) have been cleared and have recovered naturally; (ii) are at least 10 years old; and (iii) meet the lower bound of the forest threshold parameters at the start of the project.

² See Appendix 1 of the VCS Standard.

Forested wetlands, such as floodplain forests, peatland forests and mangrove forests, are also eligible provided they meet the forest definition requirements mentioned above.

ECCON utilizes the historical land use database from MapBiomas, in conjunction with deforestation and land use data from INPE, to analyze vegetation categories in each instance of the project, classified as forest by the internationally accepted definition. The definition of forest aims to verify whether the area in question corresponds to the Pantanal during the historical period of 10 years before the project's start date or inclusion in the PAI. This specific study was conducted for PAI01, and it is important to emphasize that, in the last 10 years prior to the start date, there are no records of deforestation associated with the activities of the APDef and AUDef projects, as initially mapped.

Additionally, it is important to point out that the project follows the principles outlined in section 2.2 of the VCS Standard v4.5, comprehensively meeting the requirements stipulated in section 3.2.

1.4.3 Transfer project eligibility

Not applicable.

1.5 Project Design

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

1.5.1 Grouped project design

According to the VCS Standard v4.5 (sections 3.6.17 to 3.6.18), ECCON has established the following set of eligibility criteria for the inclusion of new Project Activity Instances (PAIs).

The project zone for REDD Pantanal is defined by the geographical boundaries of the Pantanal biome, as depicted in Figure 1. These geographical limits encompass the Brazilian states of Mato Grosso and Mato Grosso do Sul.

New instances of project activities can encompass private properties along the biome, which have characteristics aligned with the additionality presented in the project. The baseline scenario is the conversion of native vegetation area for commercial purposes, in the APDef component by 20% in the forest and savannah formations of the biome. On the other hand, in the AUDef component, the scenario is the illegal conversion of areas, by 80% in the forest and savannah formations of the Pantanal biome.

In Brazil, natural vegetation is diverse and harbors complex ecosystems that are not easily defined by international bodies, such as our Cerrado vegetation. Thus, for REDD Pantanal, the UNFCCC (2006) forest definition was considered, in accordance with the Brazilian Forest Service definition of forests, to fit with the reality of our Brazilian ecosystems. The UNFCCC (2006) defines forests as an area of at least 0.05-1.0 ha with crown cover (or equivalent density) of more than 10-30%, with trees with the potential to reach a minimum height of 2-5 meters at maturity in situ. A forest can consist of both closed (dense) forest formations, where multi-strata and suppressed trees cover a high proportion of the ground, and open forests.

For the Pantanal, we consider all vegetation defined as forest by the government classification (IBGE) and all savanna vegetation that meets the UNFCCC requirements. The start date of new PAIs will be defined with the signature of the "Letter of Intent" between ECCON and landowners, as this document ensures the responsibilities for compliance with activities for participation in REDD Pantanal.

All new project instances will be described, indicating technical, financial, geographical, and other relevant information in the monitoring report during their respective inclusion periods. Additionally, they will use the same technologies and measures described in the validated PD. Activities, measures, and technologies may be adapted to the specific features of the areas, when necessary, provided that all adaptations are reported and justified in the monitoring report. According to research conducted by the project proponent on major carbon project platforms, PAI01 is not located in a jurisdictional REDD area. For each new PAI, the same assessment will be carried out to identify the existence of jurisdictional programs or other GHG programs.

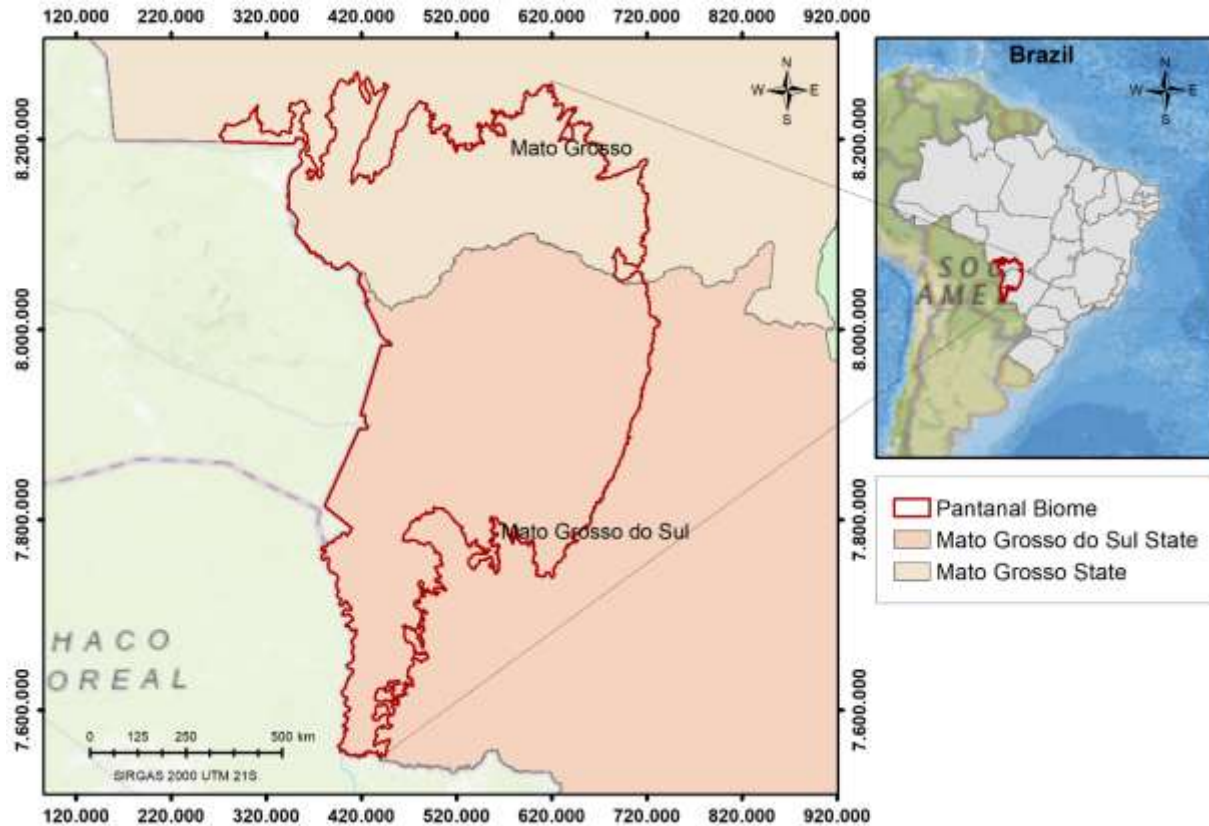


Figure 1 – Location of the Pantanal Biome in Brazil (Upper Right Panel) and the Brazil states covered by the biome (Left Panel).

1.6 Project Proponent

Organization name	ECCON SOLUÇÕES AMBIENTAIS Ltda.
Contact person	Yuri Rugai Marinho
Title	Project Developer
Address	Rua Dr. Fernandes Coelho 64 Cj 31 Pinheiros, São Paulo/SP, Brazil, CEP 05423-911
Telephone	+55 11 97603-2514
Email	yuri@ecconsa.com.br

1.7 Other Entities Involved in the Project

Not included in draft project description. Will be available at validation.

Organization name	
Role in the project	
Contact person	
Title	
Address	
Telephone	
Email	

1.8 Ownership

The Project activity will initially take place in one area in Corumbá and Miranda/MS. In accordance with the VCS Program requirements, the project ownership belongs to FAZENDA BODOQUENA LTDA ("Landowner"). On November 2021, the Landowner signed a Letter of Intent with ECCON, by which they have agreed with the elaboration and fully implementation of a REDD+ Project in the area.

Therefore, ECCON is the proponent of the Project, while the Landowner is the owner of the properties and the forests.

1.9 Project Start Date

Project start date	18-November-2021
Justification	The start of the project is 18-November-2021, according to the Letter of Intent between ECCON and Landowner.

1.10 Project Crediting Period

Crediting period	☐ <i>Seven years, twice renewable</i> ☐ <i>Ten years, fixed</i> ☒ <i>Other</i>
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Start and end date of first or fixed crediting period	18-November-2021 to 17- November -2066
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1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
18-November-2021 to 31-December-2022	34.651,77
01-January-2023 to 31-December-2023	37.785,71
01-January-2024 to 31-December-2024	43.858,33
01-January-2025 to 31-December-2025	49.928,16
01-January-2026 to 31-December-2026	55.995,21
01-January-2027 to 31-December-2027	62.059,51
01-January-2028 to 31-December-2028	68.121,08
01-January-2029 to 31-December-2029	74.179,94
01-January-2030 to 31-December-2030	80.236,12
01-January-2031 to 31-December-2031	86.289,63
01-January-2032 to 31-December-2032	86.217,71
01-January-2033 to 31-December-2033	86.146,38

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-January-2034 to 31-December-2034	86.075,61
01-January-2035 to 31-December-2035	86.005,40
01-January-2036 to 31-December-2036	85.935,76
01-January-2037 to 31-December-2037	85.866,68
01-January-2038 to 31-December-2038	85.798,15
01-January-2039 to 31-December-2039	85.730,17
01-January-2040 to 31-December-2040	85.662,73
01-January-2041 to 31-December-2041	85.595,83
01-January-2042 to 31-December-2042	85.303,50
01-January-2043 to 31-December-2043	56.795,61
01-January-2044 to 31-December-2044	51.006,67
01-January-2045 to 31-December-2045	45.218,26
01-January-2046 to 31-December-2046	39.430,38
01-January-2047 to 31-December-2047	33.643,01
01-January-2048 to 31-December-2048	27.856,15
01-January-2049 to 31-December-2049	22.069,81
01-January-2050 to 31-December-2050	16.283,97

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-January-2051 to 27-November-2051	10.498,63
Total estimated ERRs during the first or fixed crediting period	1.837.304
Total number of years	30
Average annual ERRs	61.243

1.12 Description of the Project Activity

The REDD Pantanal promotes the conservation of forests located in the Pantanal biome. This conservation will take place on private properties where there are undisputed land rights, and land tenure is resolved.

The project activities will focus on avoiding the conversion of forests (both APDef and AUDef), providing significant environmental benefits such as maintaining habitat for biodiversity, contributing to water quality through the conservation of riparian areas, and maintaining the seasonal flooding regime, thus contributing to climate change mitigation.

The project aims to mitigate the effects of climate change by reducing GHG emissions from deforestation and forest degradation, adopting measures to tackle deforestation and forest degradation factors, and reduce the loss of forest cover. Expected results include a decrease in illegal logging, restoration of degraded forests, reduction of forest conversion to other land uses, and maintenance of forest carbon stocks over time.

By promoting the conservation of native forests, the REDD Pantanal project will contribute to the conservation of forest connectivity protection, as well as structure, associated composition, functional attributes, and high conservation value, compared to baseline conditions.

To ensure the provision of the mentioned benefits, the project activities to be developed and implemented include managing and improving conservation systems for native forests and their biodiversity; supporting land use planning, land legalization, and mechanisms to ensure the sustainable development of the communities involved; contributing to technological development and innovation for the monitoring and conservation of forests and biodiversity present in the Pantanal biome.

The expected GHG reductions from the project will change over time with the inclusion of other PAIs, but considering initially PAI 01, the project's avoided emissions correspond to an annual average of 61,243 tCO₂e; therefore, over the lifespan of the project, 45 years, it will potentially generate emission reductions of up to 1.837.304 tCO₂eq (not accounting for leakage and buffer).

ECCON, as the project proponent, has a management and promotion role in conservation activities. Its main responsibilities are listed below:

- Management of REDD Pantanal
- Inventory of carbon stocks
- Monitoring of deforestation (remote sensing and in-situ visits)
- Prospecting of farms, due diligence, and contract signing
- Engagement with local authorities

The landowners will join the project and have an active role in conservation. Their main responsibilities are listed below:

- Voluntarily participate in the project
- Sign a 30-year legally binding agreement
- Halt any planned deforestation activity
- Implement measures to avoid any unplanned deforestation (active monitoring)
- Implement measures to avoid fire and degradation (e.g. firebreaks, fences)

Analyzing both the current context and the business-as-usual scenario in comparison to the implementation of REDD Pantanal, it is crucial to highlight the positive impacts resulting from the project's execution. The main pillar of these impacts lies in preventing deforestation in the project's covered areas, accompanied by a significant strengthening of governance in private areas. This contrasts with a scenario of increased deforestation in these areas, changes in water quality, modifications in the water regime, biodiversity loss and a potential increase in the frequency of extreme events, among other adverse outcomes.

The anticipated climate benefits of the project were designed using the so-called Change Theory. Table 1 illustrates the change in the causal model that explains how its activities will be carried out. The basic outline of the project, with its phases, activities, outputs, outcomes, and impacts, is presented here. The table below was created using the SBIA Guidance (Richards, M., and Panfil, S.N. 2011. Social and Biodiversity Impact Assessment (SBIA) Manual for REDD+ Projects: Part 1 – Core Guidance for Project Proponents. Climate, Community & Biodiversity Alliance, Forest Trends, Fauna & Flora International, and Rainforest Alliance. Washington, DC.) to establish a theory of change.

Table 1. Theory of change of the REDD Pantanal

Activity description	Expected climate			Relevance to project's objectives
	Outputs (short term)	Outcomes (medium term)	Impacts (long term)	
Management				
Signing of 30-year conservation agreements (ECCON+ Landowners)	One 30-year contract signed for each PAI	Increased forest protection, as conservation generates returns for landowners	Improved carbon stocks in protected forests, contributing to climate change mitigation, habitat for biodiversity and improved water quality where PAIs include riparian forest areas. Project enables environmental outcomes with communities	Very high. Project stimulates conservation through long term commitments of forest protection, generating cashflow from conservation. This improves and stimulates conservation activities, which contribute to climate change mitigation, through deforestation reduction, maintenance of habitat for fauna and flora.
Forest monitoring for degradation and deforestation (ECCON)	Annual monitoring of project areas within PAIs	Reduction of deforestation pressure and possibility of actively mitigating risks		
Measurement of biomass through forest inventory (ECCON)	Inventory plot s according to methodology for each PAI	Better estimates of carbon stock in PAIs		
Introduction of voluntary fire brigades	Training of local actors on the importance of firefighting and techniques	Improved interaction among local actors, and increased efficiency in firefighting. Improved and more resilient communities.	Improved control of wildfires, resulting in lower incidence of widespread and uncontrolled fires. Community participation in firefighting strengthens the community.	

1.13 Project Location

The REDD Pantanal is a grouped project of multiple activities located in Brazil, in the Pantanal biome. PAI 01, Bodoquena Farm, comprises the main portion of the property, located southeast of Corumbá, and a smaller portion located northwest of Miranda, both in the state of Mato Grosso do Sul, approximately 450 and 285 km away, respectively, from the state capital, Campo Grande. Bodoquena Farm has a total area of 40,408 hectares (Figure 2).

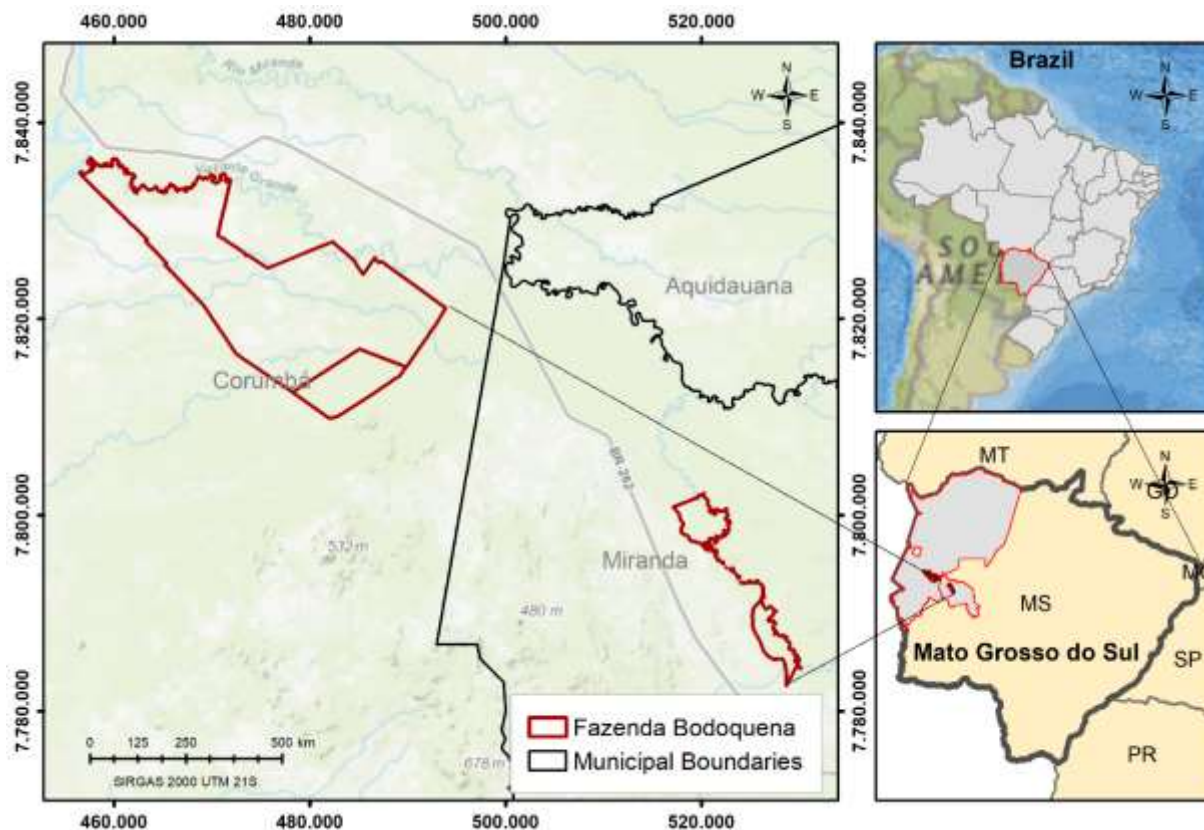


Figure 2. Illustration of the PAI. The left panel shows the boundaries of Bodoquena Farm (PAI01). The upper right panel shows the location of the state of Mato Grosso do Sul in Brazil. The lower right panel shows the boundaries of Corumbá and Miranda, with the projected location of the property indicated.

1.14 Conditions Prior to Project Initiation

According to the VM0007 methodology, AFOLU projects that include a preliminary project description for listing in the pipeline during development address the items described below:

- Ecosystem type: The project zone is classified as the Pantanal biome, a region covered by savannah formations and seasonal forests (about 31% of the biome), non-forest natural formations, wetlands, and grass formations (about 49% of the biome), agriculture and livestock (about 15% of

the biome), and water bodies (4.5% of the biome)³. PAI 01 comprises 93.4% savannah formations and 6.6% seasonal forests. The Pantanal has a typical tropical climate, classified as "Aw" according to the Köppen classification. This climate is characterized by two well-defined seasons: a dry season in winter and a rainy season in summer, with an average annual precipitation ranging between 1,000 and 1,250 mm. The average annual temperature in the region is approximately 25°C, with an average relative humidity of 82%.

- Current and historical land use: The project area has native vegetation cover, with part of the area consolidated for human use. Given the recent increase in deforestation in the region, parts of the project area are increasingly likely to be cleared and converted to pasture and cropland by smallholder farmers. In PAI 01, native vegetation is under the threat of deforestation, whether planned (APDef) or unplanned (AUDef).

- Has the land been cleared of native ecosystems within 10 years of the project start date?

☐ Yes ☒ No

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

STATUTORY AND CUSTOMARY PROPERTY RIGHTS

The areas from the project are distributed in a mosaic configuration, located in private properties, who are engaged to the Project through the Letter of Intent signed with the project proponent.

Also, in the preliminary analysis carried on by the project proponent, all the documents related to property rights were provided. In Brazil, the main ownership documents are:

Registry (Matrícula) or Ownership Term (Termo de Posse): the Registry Office provides these documents, which describe relevant historical information with regards to legal, judicial as well as financial transactions pertaining to the property (real state). If those documents reveal any legal matters, a due diligence and risk analysis is carried out.

Valid Certificate of Rural Land Record (CCIR) with the National Institute of Land Settlement and Agrarian Reform (INCRA) and, in some cases, a certified geo-referenced perimeter description of the area.

Certificate of regularity with the tax on rural property (Imposto sobre Propriedade Territorial Rural) (ITR).

Rural Environmental Registry (Cadastro Ambiental Rural) (CAR). CAR is a nationwide public registry, compulsory for all rural properties which aims to incorporate all information related to

³ <https://plataforma.brasil.mapbiomas.org/>

Permanent Preservation Areas and Legal Reserve, vegetation, consolidated areas, and others, to create a database for controlling, monitoring, environmental and economic planning and fighting off deforestation.

RECOGNITION OF PROPERTY RIGHTS

The property rights for each parcel of the mosaic configuration are recognized and respected. All properties involved in the project either have property titles or equivalent documents to certify and assure rights over the land.

If in the future any relocation of activities needs to be undertaken, it will take place with a written and signed agreement, that demonstrates consent of those involved and provides just and fair compensation.

FREE, PRIOR AND INFORMED CONSENT

The project will not encroach on private property other than that of owners formally linked to REDD Pantanal, community property, or government property.

If applicable, the process of obtaining free, prior, and informed consent will be obtained of those whose property rights will be or are affected by the project.

As mentioned, there is currently no expectation that the project will affect areas titled by people or entities other than the owners engaged with REDD Pantanal. However, if applicable, appropriate restitution or compensation will be allocated to any parties whose lands may be affected by the project.

For this topic, the document “Free, Prior and Informed Consent and REDD+: Guidelines and Resources”, from WWF, available from January 2014, was consulted.

Free, Prior and Informed Consent (“FPIC”) is a key component for the effectiveness of the Project. In the United Nations Declaration on The Rights of Indigenous Peoples, FPIC is defined as the right of indigenous peoples and traditional communities to determine and develop priorities and strategies for the development or use of their lands or territories and other resources. So, in order to obtain approval from indigenous peoples to develop a project (AFOLU, REDD, among others), it is necessary to inform and clarify all the aspects regarding to implementation, development and monitoring the project.

The 10 elements of FPIC that were considered by the Project proponent are as follows:

1. Identifying customary lands and rights holders. Process conducted by mapping of the key area.

2. Identifying and engaging with appropriate community decision-making institutions/ authorities. Process conducted by mapping of the key area and online/in-field research.
3. Identifying and engaging support organizations.
4. Building mutual understanding and agreement on a locally appropriate FPIC process.
5. Providing information. The information was provided by documentation and meetings.
6. Engaging in negotiation and supporting decision-making. Several meetings were held to support decision-making.
7. Documenting consent-based agreements. All the agreements are written and signed by the involved parties.
8. Supporting and monitoring implementation of agreements. The Project proponent has a specialized team to provide support in the implementation and monitoring of agreements.
9. Establishing and operating a conflict resolution mechanism. Communication is made via e-mail, phone or message applications, intending to guarantee an efficient conflict resolution mechanism.
10. Verifying Consent. In the Verification stage of the auditing, the documented consent will be verified by the third party.

In the proposed project, no property rights will be affected, since the project only impacts areas possessed by landowners who have already formally committed to the project through Letter of Intent. For the landowners, the process of obtaining consent was: (i) approach with landowners with explanations from an AFOLU project; (ii) several meetings for clarifications and negotiation; (iii) in-field trips and analysis provided by the project proponent; (iv) signature of agreement between the involved parts.

In the Project Zone, the communities related to the Project have been consulted by the process of Stakeholder Consultation, being done the following steps: (I) Identifying and engaging with appropriate community decision-making institutions/ authorities. Process conducted by mapping of the key area and online/in-field research; (II) Identifying and engaging support organizations; (III) Building mutual understanding and agreement; (IV) Providing information. The information was provided by documentation and meetings; (V) Engaging in negotiation and supporting decision-making. Several meetings were held to support decision-making; (VI) Establishing and operating a conflict resolution mechanism. Communication is made via e-mail, phone, or message applications, intending to guarantee an efficient conflict resolution mechanism; and (VII) Verifying Consent. In the Verification stage of the auditing, the documented consent will be verified by the third party.

PROPERTY RIGHTS PROTECTION

The project activities do not lead to any type of involuntary relocation as all the owners legally own their land and present documents that accredit it. Thus, the signed pre-contract ensure that landowners are willing to implement the project and are not being forced to relocate activities that are important to the culture or livelihood.

ILLEGAL ACTIVITY IDENTIFICATION

Illegal deforestation may affect the project. Hence, the project proponent will train local villagers to work as monitoring staff in the Project Area. This is the main activity to prevent and avoid illegal activities. The community near the Project Area will also be encouraged to report illegal loggers in the region.

The Project proponent will also monitor the occurrence of illegal activities inside the Project area. If any illegal activity is identified, a mitigation plan will be prepared, defining short, medium, and long-term strategies to address the risk to the Project, in a way to guarantee the benefits to the climate, community and biodiversity over the lifetime of the project.

ONGOING DISPUTES

The project area has no ongoing disputes. By consulting land deeds and personal documents, it is possible to verify that there are no legal issues in terms of civil, criminal, and administrative law related to the property or its owners. ECCON undertook due diligence with the information of the landowners and the farm and did not identify any disputes over lands in the past 20 years.

NATIONAL AND LOCAL LAWS

Considering that, in a wide perspective, two of the most relevant biomes in Brazil are included in the project, Amazônia and Cerrado, the importance of implementing the project is clear. They are the two largest biomes in the country, occupying an area of 49.59% and 23.92% of the territorial extension of the country, respectively. Besides that, the regulatory framework in Brazil brings specificities in the applicable rules to these biomes, given their importance.

There are lots of particularities between Cerrado and Amazônia, but we can list, specially, that (i) Amazônia biome keeps the most important Tropical Forest in the world; (ii) Amazon Forest regulates precipitation and climate in the country; (iii) Cerrado is currently the most threatened biome in Brazil, in regard to agriculture and livestock, since the expansion of these activities are, in the recent years, practically made only in Cerrado, since it has only 54.4% of its original vegetation; and (iv) accounts for the water in almost 70% of Brazil's hydrographic basins.

Still considering a wider overview, the program is also in compliance with Brazilian national laws including the Brazilian Constitution. The Chapter 6 of the Brazilian Constitution specifically

discusses environmental issues in Article 225, and the project activity aligns with the national mandate as expressed in Article 225 paragraph 4, below.

Paragraph 4 - The Brazilian Amazonian Forest, the Atlantic Forest, the Serra do Mar, the Pantanal Mato-Grossense and the coastal zone are part of the national patrimony, and they shall be used, as provided by law, under conditions which ensure the preservation of the environment.

This project is also in absolute alignment with the National Environmental Policy Law (Law No6,938 /1981), considering general national objectives of conservation and recovery of ecosystem services and many other major principles provided by the law.

Now, specifically with other legal statutes, the project details its role and impact on the Brazilian legislation.

National Environmental Laws

Law on the Protection of Native Forests - “Forest Code” (Law No. 12,651/2012): This law establishes that every rural landowner must preserve native vegetation in a portion of their property as a Legal Reserve. This portion depends on the region and biome in which the property is located. In the Amazon the Legal Reserve must be at least 80% of the property area, while in Cerrado (within the Legal Amazon) it must be 35%, and in other biomes 20%. This law also defines Areas of Permanent Protection: areas which include Riparian Zones to protect rivers, lakes, and water springs; slopes with declivities greater than 45°; “restinga” vegetation; mangroves; edges of boards and plateaus; the top of hills higher than 100 meters that have an average slope above 25°; areas with altitudes higher than 1,800 meters, regardless of the type of vegetation; and the marginal zone of veredas vegetation. The Forest Code, finally, defined the carbon credit as “a title to a tradable intangible and incorporeal asset” (Article 3, XXVII).

National Policy on Climate Change (Law No. 12,187/2009): Law No. 12,187 - Institutes the National Policy on Climate Change (Política Nacional sobre Mudança do Clima – PNMC, in Portuguese). The PNMC defines actions and measures aimed at mitigating as well as adapting to climate change, with the following specific objectives about conversion: i) Seek a sustained reduction in conversion rates, in their four-year average, in all Brazilian biomes, until zero illegal conversion is reached; ii) Eliminate the net loss of the forest cover area in Brazil, by 2020.

National Fund on Climate Change (Law No. 12,114/2009): Law No. 12,114 of December 9, 2009 - Creates the National Fund on Climate Change (Fundo Nacional sobre Mudança do Clima – FNMC, in Portuguese).

Payment for Ecosystem Services Law (Law No. 14,119/2021): Brazil has recently passed a Law for Payment for Environmental Services (nº 14.119/2021), and some articles of the law establish a foundational legal framework that gives legal security to many types of projects, as for example,

REDD+ projects, since it gives good definitions in topics that are aligned with forests, ecosystemic services and carbon credits. Native vegetation protection activities are modalities of environmental services, defined by the Law as a set of individual or collective activities that favour the maintenance, recovery or improvement of ecosystem services. REDD+ projects, for example, consist of a type of project in which there is payment for the environmental service of forest protection provided by certain individuals or communities.

State Laws

The first project area of implementation is inside the territorial extension of the state of Mato Grosso do Sul and is under these state laws and regulatory framework. Specifically, these include:

Law No. 4,163/2012: Disciplines, within the scope of the State of Mato Grosso do Sul, the exploitation of forests and other forms of native vegetation, the use of forest raw material and the obligation of reforestation. Forests and other forms of native vegetation existing in the State of Mato Grosso do Sul are considered assets of common interest to all citizens, and requirements regulated by this law for types of environmental suppression must be followed.

Law No. 4,555/2014. Creates the State Policy for Climate Change - PEMC, in the State of Mato Grosso do Sul. Establishes principles, definitions, objectives, and guidelines for the required adaptations to the impacts derived from climate change, as well as contribute to reduce or stabilize the concentration of greenhouse gases in the atmosphere.

Law No. 5,235/2018. Defines concepts, objectives, guidelines, and actions of the State Policy for Preservation of Environmental Services, establishes the State Program of Payment for Environmental Services (PESA), and establishes a Management System for this Program, in order to promote sustainable development, environmental conservation, and to encourage the provision and maintenance of these services throughout the state.

Law No. 3,020/2005. Establishes policy and norms for carbon sequestration in the state of Mato Grosso do Sul. The fixed sources of greenhouse gas emitters, especially carbon dioxide and methane, located in the State's territory, must be considered as priority environmental opportunities for carbon projects and their respective credits, to be monetized by the Clean Development Mechanism of the Kyoto Treaty, as well as other existing markets.

Decree Nº. 13,777/2013. Determines that the State of Mato Grosso do Sul, by means of eligibility criteria, will recognize sustainable development programs that aim the preservation of native forests, associated with the promotion of agricultural production, cattle-raising, and sustainable forestry, and that can generate voluntary socio-environmental and cultural compensation bonds.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes

☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes

☐ No

1.18 Sustainable Development Contributions

The REDD Pantanal project aims to reduce deforestation in vulnerable areas, thereby contributing to the decrease in greenhouse gas emissions from avoided deforestation and the protection of native vegetation. These actions will result in a series of benefits for the climate, local communities, and biodiversity, ensuring that the project's initiatives align with the United Nations' Sustainable Development Goals (SDGs). Brazil formally committed to these goals during the Agenda 2030 conferences and by participating in the Paris Agreement in 2015.

The project activities seek to contribute to the Sustainable Development Goals (SDGs) by addressing deforestation prevention and ensuring the protection of soil and water resources in the project areas. Additionally, they aim to promote sustainable production, as revenue from the sale of carbon credits will incentivize landowners to improve conservation practices and keep native vegetation protected.

The anticipated outcomes of these activities include the project's contribution to climate change mitigation by avoiding greenhouse gas emissions from deforestation, thus ensuring a stable local climate. Furthermore, the project aims to protect biomes that are vulnerable to deforestation and are designated as Priority Areas for Conservation, as outlined in National Decree 5,092/2004).

To ensure the project functions as planned, REDD Pantanal will utilize satellite imagery and on-site visits to assess conservation in project areas. This approach helps protect the environment by preventing deforestation and maintaining the quality of soil, rocks, and water. It also contributes to maintaining groundwater levels and river flow, crucial for local communities.

- The Project Zone is in the Pantanal biome, which has faced challenges with deforestation in recent years. In 2022, the deforestation rate in the Amazon was 31,211.7 hectares. Between 2019 and 2022, there was a cumulative deforestation of 101,064 hectares, with an average annual increase of 31% during this period.

The biome is at risk of illegal and legal conversion, being part of national efforts to reduce deforestation. Elevated levels of deforestation may push the biome beyond a critical point, threatening its ecological functions, climate regulation, food security for communities, and development capacity. The REDD Pantanal project aims to reduce deforestation pressure in these areas by preserving endangered vegetation.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Not included in draft project description. Will be available at validation.

1.19.2 Commercially Sensitive Information

Not included in draft project description. Will be available at validation.

1.19.3 Further Information

Not included in draft project description. Will be available at validation.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Not included in draft project description. Will be available at validation.

2.1.2 Stakeholder Consultation and Ongoing Communication

Not included in draft project description. Will be available at validation.

2.1.3 Free Prior and Informed Consent

Not included in draft project description. Will be available at validation.

2.1.4 Grievance Redress Procedure

Not included in draft project description. Will be available at validation.

2.1.5 Public Comments

Not included in draft project description. Will be available at validation.

2.2 Risks to Stakeholders and the Environment

Not included in draft project description. Will be available at validation.

2.3 Respect for Human Rights and Equity

Not included in draft project description. Will be available at validation.

2.3.1 Labor and Work

Not included in draft project description. Will be available at validation.

2.3.2 Human Rights

Not included in draft project description. Will be available at validation.

2.3.3 Indigenous Peoples and Cultural Heritage

Not included in draft project description. Will be available at validation.

2.3.4 Property Rights

Not included in draft project description. Will be available at validation.

2.3.5 Benefit Sharing

Not included in draft project description. Will be available at validation.

2.4 Ecosystem Health

Not included in draft project description. Will be available at validation.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF),	1.6

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Module	VMD0001	VMD0001 Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools (CP-AB)	1.1
Module	VMD0002	VMD0002 Estimation of carbon stocks in the dead-wood pool (CP-D)	1.1
Module	VMD0004	VMD0004 Estimation of stocks in the soil organic carbon pool (CP-S)	1.0
Module	VMD0006	VMD0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation and planned degradation (BL-PL)	1.3
Module	VMD0007	VMD0007 Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation (BL-UP)	3.3
Module	VMD0009	VMD0009 Estimation of emissions from activity shifting for avoided planned deforestation (LK-ASP)	1.3
Module	VMD0010	VMD0010 Estimation of emissions from activity shifting for avoided unplanned deforestation (LK-ASU)	1.2
Module	VMD0013	VMD0013 Estimation of greenhouse gas emissions from biomass burning (E-BPB)	1.2
Module	VMD0015	VMD0015 Methods for monitoring of greenhouse gas emissions and removals (M-REDD)	2.2
Module	VMD0016	VMD0016 Methods for stratification of the project area (X-STR)	1.2

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Module	VMD0017	VMD0017 Estimation of uncertainty for REDD project activities (X-UNC)	2.2
Tool	VT0001	VT0001 Tool for the demonstration and assessment of additionality in VCS agriculture, forestry, and other land use (AFOLU) project activities (T-ADD)	3.0
Tool	-	AFOLU Non-Permanence Risk Tool	4.2
Tool	-	CDM Executive Board "Tool for testing significance of GH G emissions in A/R CDM project activities	1.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0007	Land in the project area has qualified as forest (following the definition used by VCS) for at least 10 years before the project start date.	The project area is qualified and meets the condition, with ecosystem functioning as a forest in the period between 2012 and 2021.
VM0007	Baseline deforestation in the project area falls within either of the following categories: a) Unplanned deforestation (VCS category AUDef) b) Planned deforestation (VCS category APDef)	The baseline deforestation and forest degradation in the project area fall within the Categories: Unplanned and planned deforestation (AUDef and APDef categories).

Methodology ID	Applicability condition	Justification of compliance
VMD0001	This module is applicable to all forest types and age classes.	The module is applicable to the uneven-aged forest types of the project.
VMD0002	This module is applicable to all forest types and age classes.	The module is applicable to the uneven-aged forest types of the project.
VMD0004	This module is applicable to non-organic soils under all forest types and age classes.	The module is applicable to non-organic soils and the uneven-aged forest types of the project.
VMD0006	The module is applicable for estimating the baseline emissions on forest lands (usually privately or government owned) that are legally authorized and documented to be converted to non-forest land.	Project activities involve avoiding planned deforestation (APDef), so the module is applicable. Legal deforestation in the country is regulated by the Brazilian Forest Code (Law No. 12,651/2012) and associated state legislation. There are no legal limits of which proportion of the farm can be legally deforested each year.
VMD0006	Where, pre-project, unsustainable fuelwood collection is occurring within the project boundaries Modules BL-DFW and LK-DFW must be used to determine potential leakage.	Condition not applicable. There is no history of fuelwood collection identified within the project boundary, this activity is excluded from the baseline and project.
VMD0007	The module must be applied to all project activities where the baseline agents of	In the basic scenario, deforestation agents convert land to livestock and small-scale agricultural

Methodology ID	Applicability condition	Justification of compliance
	deforestation: (i) clear the land for settlements, crop production (agriculturalist), ranching or aquaculture, where such clearing for crop production, ranching or aquaculture does not amount to large scale industrial agri/aquaculture activities; (ii) have no documented and uncontested legal right to deforest the land for these purposes; and (iii) are either resident in the region or immigrants.	production, where there is no legal right to clear the land for these purposes; agents are local residents of the region or immigrants seeking land for land use as pasture.
VMD0007	Where pre-project, unsustainable fuelwood collection is occurring within the project boundaries, Modules BL-DFW and LK-DFW must be used to determine potential leakage.	Condition not applicable. There is no history of fuelwood collection identified within the project boundary, this activity is excluded from the baseline and project.
VMD009	The module is applicable for estimating the leakage emissions due to activity shifting from forest lands that are legally authorized and documented to be converted to non-forest land, including activity shifting to forested wetland that is drained or degraded as a consequence of project implementation.	The activities of the project involve avoiding planned deforestation (APD), because the project area has legal permissibility for land conversion, so that the module is applicable.

Methodology ID	Applicability condition	Justification of compliance
	The module is also applicable for estimating the leakage emissions due to activity shifting from non-forested wetlands that are legally authorized and documented to be converted and degraded.	
VMD0009	This tool must be used for projects in areas where planned deforestation happens on forested wetlands, regardless of the absence of wetland within the project boundaries.	The module is applicable to wetlands in the project area.
VMD0009	The module is mandatory if Module BL-PL has been used to define the baseline, and the applicability conditions in Module BL-PL must be complied with in full.	The BL-PL module is applied to the project and meets the established applicability conditions.
VMD0010	This module is applicable for estimating carbon stock changes and greenhouse gas emissions related to the displacement of activities that cause deforestation of lands or wetland degradation outside the project area due to avoiding unplanned deforestation or avoiding unplanned wetland degradation in the project area.	The activities of the project involve avoiding unplanned deforestation (AUDD), so the displacement of these activities should be monitored, since the pressure for deforestation in the project zone includes the conversion of forest land into to grazing lands and crop lands.

Methodology ID	Applicability condition	Justification of compliance
	Activities subject to potential displacement are conversion of forest land to grazing lands, crop lands, and other land uses, or conversion of intact or partially degraded wetlands to drained or degraded wetlands.	
VMD0010	The module is mandatory if module BL-UP has been used to define the baseline and the applicability conditions in module BL-UP must be complied with in full.	The BL-UP module is applied to the project and meets the established applicability conditions.
VMD0013	This module is applicable to REDD project activities with emissions from biomass burning and REDD-WRC project activities with emissions from biomass and/or peat burning. This module is also applicable to RWE and ARR-RWE project activities with emissions from peat burning.	The module is applicable to all atividades de projetos de REDD com emissões provenientes de queima de biomassa.
VMD0015	The module is mandatory for REDD, CIW-REDD, RWE-REDD, and stand-alone CIW project activities. Where selective logging is taking place in the project case:	The module is mandatory. In addition, the stratification is fixed ex-ante for the baseline and will not be changed and there are no selective logging activities taking place in the project area.

Methodology ID	Applicability condition	Justification of compliance
	• Emissions from logging may be omitted if it can be demonstrated the emissions are de minimis using tool T-SIG or Appendix 1 of REDD-MF. • If emissions from logging are not omitted as de minimis, logging may only take place within forest management areas that possess and maintain a Forest Stewardship Council (FSC) certificate for the years when the selective logging occurs. • Logging operations may only conduct selective logging that maintains a land cover that meets the definition of forest within the project boundary. • All trees cut for timber extraction during logging operations must have a DBH greater than 30 cm. • During logging operations, only the bole/log of the felled tree may be removed. The top/crown of the tree must remain within the forested area. • The logging practices cannot include the piling	The other conditions do not apply.

Methodology ID	Applicability condition	Justification of compliance
	and/or burning of logging slash • Volume of timber harvested must be measured and monitored.	
VMD0016	For peatland rewetting and conservation project activities this module must be used to delineate non-peat versus peat and to stratify the peat according to peat depth and soil emission characteristics, unless it can be demonstrated that the expected emissions from the soil organic carbon pool or change in the soil organic carbon pool in the project scenario is de minimis, In the case of WRC project activities, the project boundary must be designed such that the negative effect of drainage activities that occur outside the project area on the project GHG benefits are minimized.	The module is mandatory. The module is applicable due to the variation of classes of Brazilian forests, considering the different phytophysiognomies as strata i.
VMD0017	This module is mandatory when using methodology REDD+ MF. It is applicable for estimating the uncertainty of estimates of emissions and removals of	This module is mandatory when using methodology REDD+ MF, which is applied to the project.

Methodology ID	Applicability condition	Justification of compliance
	CO2-e generated from REDD and WRC project activities.	

3.3 Project Boundary

Not included in draft project description. Will be available at validation.

3.4 Baseline Scenario

Not included in draft project description. Will be available at validation.

3.5 Additionality

Not included in draft project description. Will be available at validation.

3.6 Methodology Deviations

Not included in draft project description. Will be available at validation.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Not included in draft project description. Will be available at validation.

4.2 Project Emissions

Not included in draft project description. Will be available at validation.

4.3 Leakage Emissions

Not included in draft project description. Will be available at validation.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Not included in draft project description. Will be available at validation.

5 MONITORING

5.1 Data and Parameters Available at Validation

Not included in draft project description. Will be available at validation.

5.2 Data and Parameters Monitored

Not included in draft project description. Will be available at validation.

5.3 Monitoring Plan

Not included in draft project description. Will be available at validation.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

